

⚠ REPRODUCTION OF THIS MATERIAL IS STRICTLY PROHIBITED

Find the area between the curve

$$y = \sin^2(ax)$$

and the  $x$ -axis between the bounds  $x = 0$ , and  $x = \frac{\pi}{a}$ .

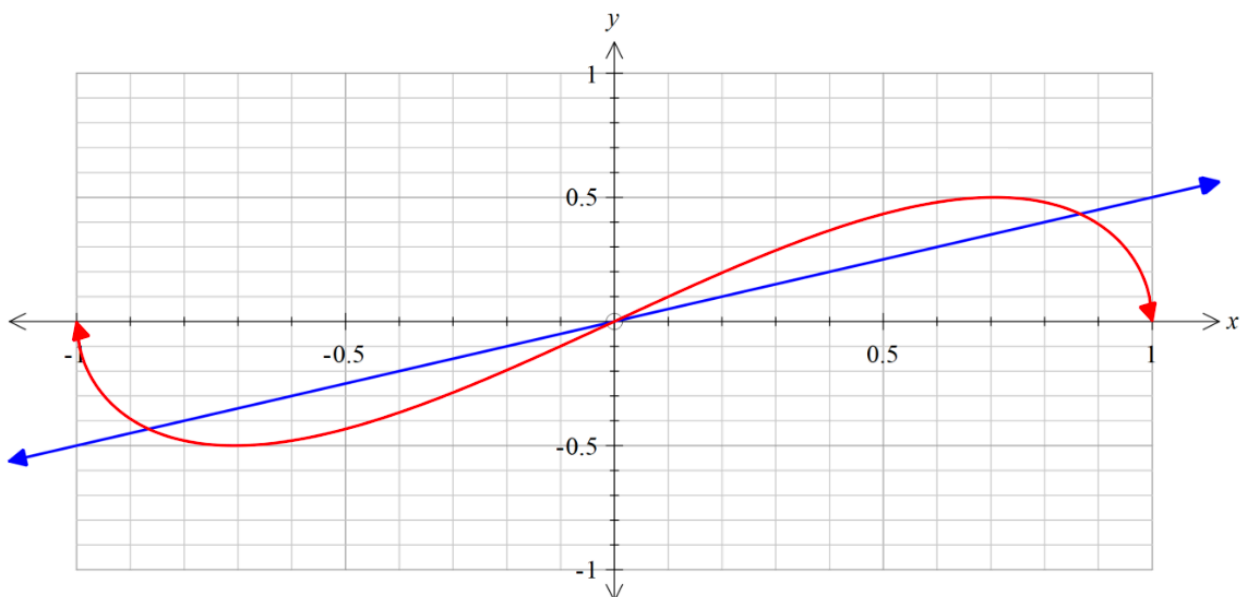
Frank is feeling unwell. His wife Mary makes him a hot lemon drink to make him feel better and takes it to him in a room that is  $17^\circ\text{C}$ .

Write the differential equation that expresses this statement.

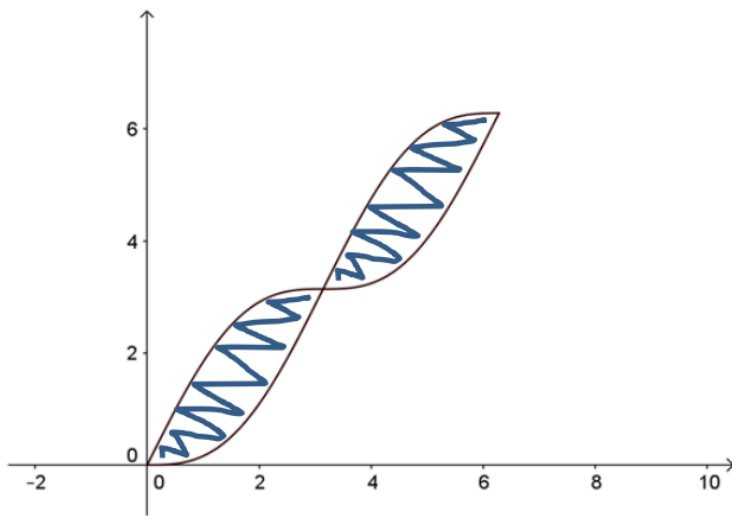
When the hot lemon drink is poured it is  $97^\circ\text{C}$ . After 3 minutes, the hot lemon drink has cooled to  $68^\circ\text{C}$ . Mary tells Frank that he must drink the hot lemon drink while it is between  $48^\circ\text{C}$  and  $56^\circ\text{C}$  if it is to help him feel better.

Solve the differential equation to find how long since it was poured, that Frank can begin drinking the hot lemon drink. Also, find how long Frank has to finish his drink.

Find the areas between the line  $y = 0.5x$  and the curve  $y = x\sqrt{1-x^2}$  shown below.



An aeronautical engineering company is attempting to re-design a propeller for a helicopter. Using the graph below to work out the area of the 2-dimensional representation of their design.



The two functions are

$$f(x) = x + \sin x$$

and

$$g(x) = x - \sin x$$

An integration technique called 'Integration by Parts' allows some products to be integrated. These products need to be in the form of one function multiplied by another function's derivative, or  $f(x)g'(x)$ . The rule for integration by parts is:

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx$$

Using this, or otherwise, find

$$\int x \sin x dx$$

A particular colony of the microscopic Sea Bear has contracted the rare disease called Mathelitis. This causes the population to decline at a rate that is proportional to the current size of the population of Sea Bears.

The population 16 days after contracting the disease was 735 and after 22 days the population had dropped only to 164.

By forming and solving a differential equation suitable to this situation, find out the initial population size and during what day it is expected that only 10 Sea Bears are left.

A space craft is accelerating after leaving orbit around a planet in a particular way such that it can be described using the function

$$a(t) = -\sqrt{t^2} + 3t^2 + \frac{1}{(t+1)^2}$$

The acceleration is measured in terms of the spacecraft motion in relation to the sun. The time is measured in months since leaving orbit, and acceleration is measured in  $\cdot 10^3 km s^{-2}$ . Given the velocity of the

spacecraft was  $-1000\text{km s}^{-1}$  when it left orbit show that spacecraft enters the orbit during the second month (assuming the direction of travel is in a straight line, either way from or towards the planet).

---

Find  $k$  s.t.

$$\int_2^4 \sqrt[3]{x-2} \, dx$$

is twice that of

$$\int_k^4 \sqrt[3]{x-2} \, dx$$

---

A cylindrical tank has radius of 1 m and height of 2.6 m a tap at the bottom of the tank is turned on and the water drains from the tank. One model involving the volume of water in the tank is to take the rate at which the volume of water in the tank changes at any instant to be proportional to the square root of the height of the water in the tank at any instant.

Initially, the tank is full of water. After 5.1 hours after the water in the tank has drained out.

For my differential equation that uses the model above and solve it to find how long it takes for all the water to drain from the tank.

---

Given

$$\int uv' \, dx = uv - \int u'v \, dx$$

evaluate

$$\int_0^1 xe^x \, dx$$

---

Find the general solution of the differential equation

$$4(x^2 + 3) \frac{dy}{dx} = x(16 - y^2)$$

If  $y = 2$  when  $x = 1$ , find the value of the constant of integration.

Hint: if

$$\frac{a}{a^2 - x^2} = \frac{P}{a + x} + \frac{Q}{a - x}$$

then  $P = Q = \frac{1}{2}$

---

Find the area enclosed between

$$y = \cos^2 x$$

and

$$y = \sin^2 x$$

from  $x = -\frac{\pi}{4}$  to  $x = \frac{\pi}{4}$ .

---

A blacksmith is making an item from steel that has a melting point of  $1370^\circ\text{C}$ . One melted, the liquid steel was poured into the mould for the object and left to cool. It is known that this object is thin enough to cool uniformly. It cools at a rate that is proportional to the difference between its current temperature and the ambient temperature of its surroundings. At  $140^\circ\text{C}$  the object is cool enough to hold its shape and can be taken out after mould. If the ambient temperature of the blacksmith workshop is  $47^\circ\text{C}$  and after 10 minutes, the object had to go to  $850^\circ\text{C}$  how long before it can be removed from its mould?

---

A curve passes through  $(1, 2)$  and  $(\frac{1}{4}, -10)$  and has a gradient which is inversely proportional to  $x^2$  (i.e. proportional to  $\frac{1}{x^2}$ ). Find the equation of the curve.

---

Find the solution to the differential equation

$$\frac{dy}{dx} = \frac{y - 4xy}{x - 2x^2}$$

where the point  $(1, 5)$  is on the solution curve.

---

A new brand of ice cream, SuperFreeze, warms up at the right that is proportional to the difference between its temperature and the temperature of the room within which it is stored. The ice cream is delivered at  $-15^\circ\text{C}$ , and five minutes later it is  $-12.5^\circ\text{C}$ . Given the storage room is a constant  $0^\circ\text{C}$  and the ice cream cannot be served colder than  $-3^\circ\text{C}$ , calculate how long after delivery the ice cream can be served.

---

The right at which of body loses velocity at any instant as it passes through a medium it's given by

$$-kv \text{ m/s}^2$$

Where  $v$  is the velocity of the body at that instant and  $k$  is a positive constant. If its initial velocity is  $u \text{ m/s}$ , show that the time taken for the body to halve its velocity is

$$\frac{\ln 2}{k}$$

seconds.

---

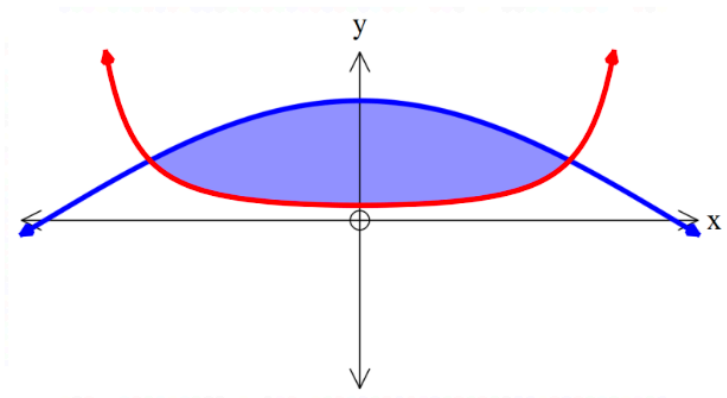
The diagram below shows the graphs of

$$y = 8 \cos(kx)$$

and

$$y = \sec^2(kx)$$

for the same constant  $k$ .



Find an expression for the shaded area bounded by the two curves.

---

The line  $y = ax$ , where  $0 < a < 5$ , intersects the curve

$$y = x(5 - x)$$

at  $x = 0$  and  $x = p$  as shown in the diagram below.

Show that the area between the line  $y = ax$  and the curve  $y = x(5 - x)$  bounded by  $x = 0$  and  $x = p$  is  $\frac{p^3}{6}$ .

---

A paper cup with a height of 10 cm is filled with water. The cup has a small hole in the bottom and water is leaking out of the cup. The rate of change of the height of water in the cup is modelled by the equation

$$y^2 \frac{dy}{dx} = -k\sqrt{y}$$

where  $y$  is the height of water in the cup in cm, and  $t$  is time in minutes.

After two minutes, the height of water in the cup is nine centimetres.

What is the height of water in the cup after eight minutes?

---

The diagram shows the graph of

$$y = x\sqrt{k - x^2}$$

Find the area in terms of  $k$  bounded by  $y = x\sqrt{k - x^2}$  and the  $x$ -axis.

---

In a chemical reaction the mass,  $M$  grams, of a chemical at time  $t$  seconds into the process is modelled by the differential equation

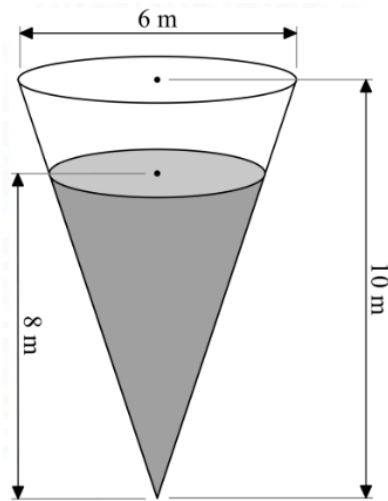
$$\frac{dM}{dt} = \frac{M}{t(1+t^2)}$$

When  $t = 1$  second the mass is  $M = 25$  grams. Find an equation for the mass  $M$  in terms of  $t$ .

Helpful hint:

$$\frac{1}{t(1+t^2)} = \frac{1}{t} - \frac{t}{1+t^2}$$

A water tank is shaped like a cone as shown in the diagram the tank has a diameter of 6 m and a height,  $H$ ,



of 10 m.

The energy required to pump water out of a tank of height,  $H$ , is related to the depth of water in the tank and the top of the surface area of the water. The amount of energy required is given by

$$E = 9800 \int_{H-d}^H (H-h)A(h) dh$$

Where  $E$  is the energy in joules  $d$  is the initial depth of the water  $h$  is the depth of the water in the tank at any instant  $A(h)$  is the area of the surface of the water at this instant.

Find the energy required to pump all the water out of the tank shown.

The variables  $y$  and  $t$  satisfy the differential equation

$$\frac{dy}{dt} = (y+2)\sin^2 t$$

Also, when  $t = 0$ ,  $y = 0$ . Find the value of  $y$  when  $t = \frac{\pi}{4}$ .

Solve the differential equation

$$(x+2)\frac{dy}{dx} = (2x^2 + 4x + 1)(y-3)$$

Hence, find an expression in the form  $y = f(x)$  given that  $y = 11$  when  $x = 0$ .

The graph below shows the curves

$$y = \sec x \tan x$$

and

$$y = \csc x \cot x$$

between  $x = 0$  and  $x = \frac{\pi}{2}$ .

Find the value of  $k$  so that the two shaded regions have equal areas.

---

The cent of mass of an object is called the centroid.

For a uniformly thin object, concentrate is that the coordinates  $(\bar{x}, \bar{y})$  given by:

$$\bar{x} = \frac{1}{A} \int_a^b x f(x) dx$$
$$\bar{y} = \frac{1}{A} \int_a^b \frac{[f(x)]^2}{2} dx$$

where  $A$  is the area of the object and  $a$  and  $b$  out lower and upper limit of  $x$  respectively.

The shape shown shaded below is bounded by the curve

$$y = \frac{1}{x}$$
$$x = 1$$
$$x = e$$

and the  $x$ -axis.

Find the coordinates of the centroid of the shape.

---

Prove that

$$\int_0^{\frac{\pi}{4}} \tan^4 x dx = \frac{\pi}{4} - \frac{2}{3}$$

---

Newtons law of cooling states that the rate of heat loss of a body is directly proportional to the difference in the temperatures between the body and its surroundings. Use this law to solve the following problem. The temperature of a room is  $25^\circ\text{C}$  an object which has the temperature of  $50^\circ\text{C}$  was placed on a table to cool after five minutes. The temperature of the object had fallen to  $40^\circ\text{C}$ . Find the time it takes for the temperature of the object to fall to  $30^\circ\text{C}$  from when it was first placed on the table.

---

The graph below shows the curves

$$y = x(x - k)$$

and

$$y = x(x - k)^2$$

where  $k$  is a positive constant.

Show that the shaded area is

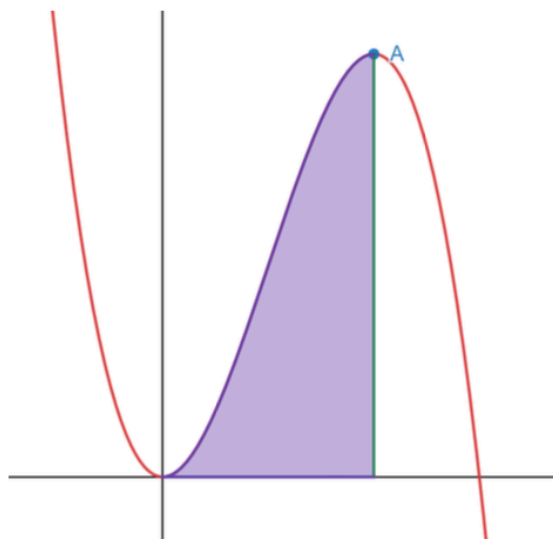
$$Ak^3(k + B)$$

AND evaluate the constants  $A$  and  $B$ .

---

The graph of

$$y = x^2(p - x)$$



is shown on the axes below.

Point

$A$  is the maximum point of the curve. Show that the shaded area is

$$\frac{4}{81}p^4$$


---

A balloon has a leak and it's slowly decreasing in volume. Let  $V \text{ cm}^3$  be the volume of the balloon after  $t$  seconds. The rate of change of the volume to balloon can be modelled by the differential equation

$$\frac{dV}{dt} = -kV^{5/3}$$

The initial volume of the balloon ( $t = 0$ ) is  $1000 \text{ cm}^3$ . After five seconds the volume of the balloon is  $512 \text{ cm}^3$ .

What is the volume of the balloon after 16 seconds?

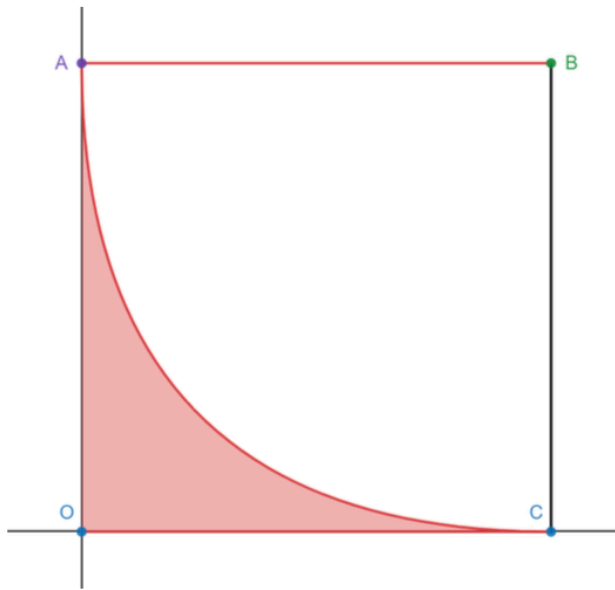
---

The graph of

$$\sqrt{x} + \sqrt{y} = k$$

where  $k > 0$  is shown on the axes below.





the area of OABC

Show that the shaded area is  $\frac{1}{6}$  of

An object motion is oscillating up and down in such a manner that its acceleration function can be modeled by

$$a(t) = 6 \cos\left(\frac{t}{5}\right) \cos\left(\frac{t}{10}\right)$$

where  $t$  is the time in seconds, and its displacement was measured in metres.

The object was initially stationary.

At 15 seconds it was noticed the object was observed to be traveling upwards during one of its oscillations. Find the vertical displacement of the object moves in the 16th second of motion.

Hot oil is used to deep fry chips on a stove top element. The oil is left a cool so that it can be returned to a plastic bottle for storage. The temperature of the oil is  $283^{\circ}\text{C}$  15 minutes after the element is turned off the temperature is  $167^{\circ}\text{C}$  45 minutes after the element was turned off. The oil can only be returned to the plastic container for storage when it has cooled down to  $40^{\circ}\text{C}$  how long after the element is turned off will the oil be able to be returned to the plastic container? The room temperature throughout the cooling process is  $22^{\circ}\text{C}$ .

This situation can be modelled using

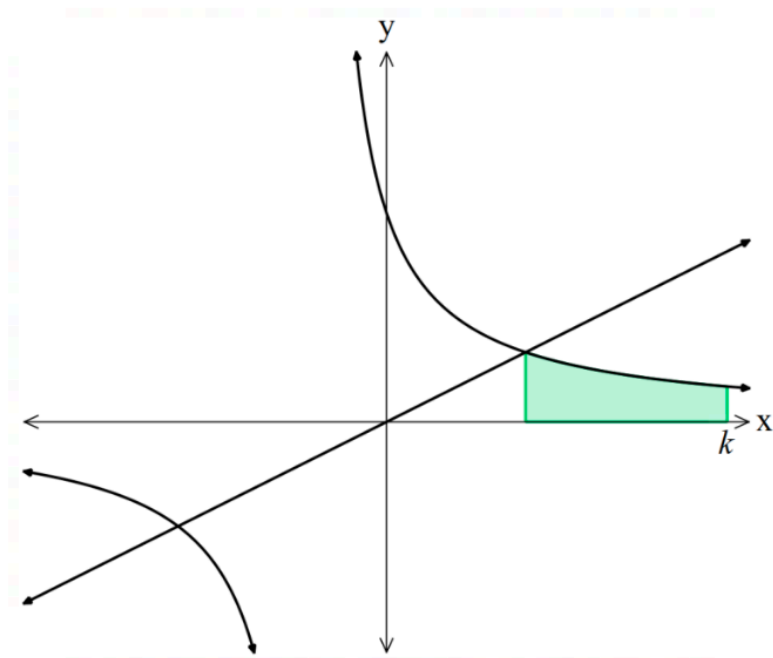
$$\frac{dT}{dt} = k(T - T_0)$$

The graphs of

$$y = \frac{3}{x+1}$$

and

$$2y = x$$



are shown on the graph below.

The

shaded region has an area of two square units. Find the value of  $k$ .

A safe laptop temperature is generally between  $0^{\circ}\text{C}$  and  $32^{\circ}\text{C}$ . One laptop overheated and reached an internal temperature of  $90^{\circ}\text{C}$ . It was then turned off to cool down in a room with a constant temperature of  $25^{\circ}\text{C}$ , and after 10 minutes the temperature drops to  $60^{\circ}\text{C}$ .

How long will it take for the laptop to cool down to a safe operating temperature?

The growth rate of the field mice population in a particular region is inversely proportional to the square root of the population  $P$ . The population eight years ago ( $t = 0$ ) was 15,000 and 5 years ago ( $t = 3$ ) was 24,000. Find the population now.

The graph

$$y = \frac{4 \sin^2 \left( \frac{x}{k} \right) - 1}{\sin^2 \left( \frac{x}{k} \right)}$$

has roots at  $x = \frac{k\pi}{6}$  and  $x = \frac{5k\pi}{6}$ .

Determined the area bounded by the graph, the  $x$ -axis, and the two roots.

A bowl of hot food is placed on a table to cool. The room temperature is constant at  $20^{\circ}\text{C}$ . The rate at which the temperature of the food cools is proportional to the difference between the temperature of the bowl and the temperature of the room. The temperature of the bowl when placed on the table was  $80^{\circ}\text{C}$ . After two minutes the temperature of the bowl was  $70^{\circ}\text{C}$ . The food will be cool enough to eat when the temperature has fallen to  $56^{\circ}\text{C}$ . How long will it be from the time the bowl is placed on the table until the food is sufficiently cool to eat?

The model for the height  $h$  metres, of a tree  $t$  months after being planted, says that while the tree is growing, the rate of increase of height with respect to time is  $k\sqrt[3]{9-h}$  metres per month. When  $t = 0$ ,  $h = 1$ , and the rate at which the tree is growing is  $0.4m$  per year. Find the time it takes for the tree to achieve its maximum height.

Find the value of  $x$  when  $y = 9$  if

$$\frac{dy}{dx} = \frac{8ye^{2x}}{1 + e^{2x}}$$

and  $y = 4$  when  $x = 0$ .

A dog's acceleration in  $\frac{m}{s^2}$  is given by  $a(t) = (kt + 1)^{-2}$ , where  $m$  is the distance in metres from a stick and  $t$  is the time in seconds. Initially, the dog's velocity is  $4\frac{m}{s}$  and it is 2 metres from the stick. Find an expression for the dog's distance from the stick after  $t$  seconds.

Find an explicit expression for  $y$  in terms of  $x$ , if

$$(x + 2)\frac{dy}{dx} = (2x^2 + 4x + 1)(y - 3)$$

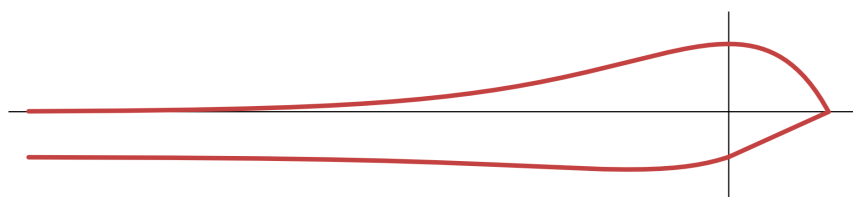
and  $y = 7$  when  $x = 0$ .

Engineers are designing a new style of aircraft wing. The shaded area below shows the latest design from a cross section view. It comprises of the following functions

$$y_1 = -0.3(x - 3)e^x, \{-7 < x < 1\}$$

$$y_2 = 0.15e^x - 0.2, \{-7 < x < 0\}$$

$$y_3 = 0.2x - 0.2, \{0 < x < 1\}$$



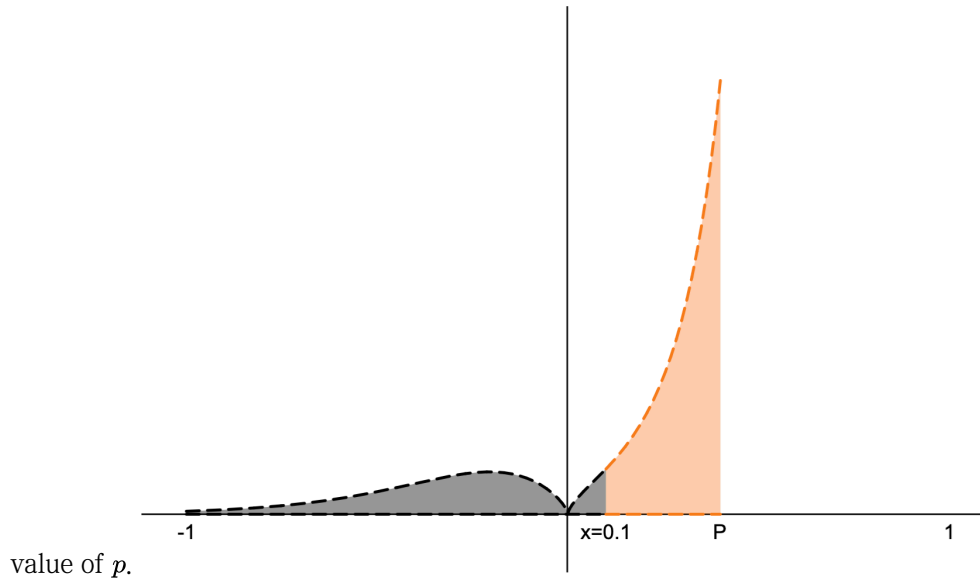
new wing design.

Find the cross sectional area of the

The following graph shows the function

$$y = x^{\frac{2}{3}} \left(1 - x^{\frac{5}{3}}\right)^{-6}$$

from  $-1 < x < p$ . The two shaded areas from  $-1 < x < 0.1$  and  $0.1 < x < p$  are equal in size. Calculate the



Find  $k$  such that the area enclosed by the lines  $x = 0$ ,  $x = 5$ , the  $x$ -axis, and the curve

$$y = e^{kx}$$

is greater than

$$\frac{5}{k}$$

Find the value of  $x$  when  $y = 9$  if

$$\frac{dy}{dx} = \frac{6ye^{3x}}{1 + e^{3x}}$$

and  $y = 4$  when  $x = 0$ .

The number of germs in a petri dish  $t$  hours after the start of observations is denoted by  $N$ . The variation in the number of germs is modelled by the differential equation

$$\frac{dN}{dt} = kN$$

where  $k$  is a constant and  $N$  is taken to be a continuous variable measured in millions. It is given that  $N = 50$  when  $t = 0$ . Given also that  $N = 100$  when  $t = 10$ , find how many hours pass before the number of germs in the petri dish has  $N$  exceeding 1000.

A patient has a glucose solution administered intravenously into his bloodstream at a constant rate,  $r$ . As the glucose is added, it is converted into other substances and removed from the bloodstream at a rate that is proportional to the concentration at the time.

A model  $C(t)$ , the concentration of the glucose solution in the bloodstream, is given by

$$\frac{dC}{dt} = r - kC$$

where  $k$  is a positive constant.

At time  $t = 0$ , the concentration is  $C_0$  where  $C_0 < \frac{r}{k}$ .

Find an expression for the concentration  $C(t)$  at any time, and explain what happens as  $t$  increases.

---

Find the area between the curve

$$y = \cos^2 x$$

and the  $x$ -axis,  $n\pi \leq x \leq (n\pi + 1.5\pi)$ .

---

$A$  is the area enclosed by the graph of the function  $y = f(x)$ , the lines  $x = a$ , and  $x = b$ , and the line  $y = c$  where  $c > f(x)$  for all  $x$  satisfying  $a < x < b$ .

Using all integrals involved, prove that

$$A = c(b - a) - \int_a^b f(x)dx$$

It may be assumed that  $f(x) > 0$  for all  $x$  satisfying  $a < x < b$ .

---

The number of radioactive particles in a substance  $t$  years after the start of observations is given by  $N$ . The variation in the number of radioactive particles is modelled by differential equation

$$\frac{dN}{dt} = kN$$

where  $k$  is a constant, and  $N$  is taken to be a continuous variable measured in billions. It is given that  $N = 25$  when  $t = 0$ .

i) Solve the differential equation, obtaining a relation between  $N$ ,  $k$ , and  $t$ . ii) Given also that  $N = 10$  when  $t = 5$ , find the value of  $k$ . Hence find the time that it takes for the number of radioactive particles to halve.

---

The height  $h$  in  $cm$ , of the fluid in a container at time  $t$  seconds, is given by the differential equation

$$\frac{dh}{dt} = \frac{A}{-\sqrt{h^3}}$$

where  $A$  is a constant. Initially, the height of the fluid is  $H$   $cm$ , and after 60 seconds, the container is empty. Find whether the time required for the height of fluid to be  $\frac{H}{2}$   $cm$  is more or less than 30 seconds, and by how much more or less.

---